Benchmarking Sepsis AI Clinical Systems Under Consent Constraints

Prediction and Alert Changes Over Time

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Abstract

Can a sepsis prediction system retain useful performance when patients restrict its inputs? We present a reproducible benchmark evaluation on all 40,336 public PhysioNet 2019 patients, comparing three supervised models across three split seeds, two hospitals and eight simulated permission conditions. Within each comparison, patients, labels, fitted predictors and alert thresholds remain fixed; only permitted information and derived state change. Paired whole-patient resampling measures differences in discrimination, incident misses and review burden. At a 30% independent denial probability for each physiological feature group, mean AUROC falls by 0.019–0.051 across model–hospital combinations. Effects depend strongly on the restricted source. At hospital B, removing laboratory inputs changes GRU AUROC from 0.752 to 0.745, yet increases missed-incident risk by 11.2 percentage points at the original hourly alert threshold while reducing reviews by 1.42 per patient. Thus similar ranking performance can conceal a substantial change in alert behavior. We formalize paired loss changes and give a bounded-loss preservation criterion with an explicit proof based on established concentration methods. Joint policy analyses evaluate alert timing, review capacity and modeled oversight alongside the paired input comparison. Hourly trajectories and simulated withdrawal at hour 12 make the monitoring-time scope explicit. The contribution is an evaluation protocol and measured failure analysis, not a new predictor or a determination of ethical acceptability. Restrictions act at inference time on models trained with full public inputs; real consent, clinical non-inferiority and patient benefit remain unmeasured.

Introduction

Clinical AI systems can gain predictive information by combining many sensitive observations. That creates a practical question: how much of their useful performance survives when patients limit which data the system may process? We examine this question through sepsis prediction, comparing full-input reference models with the same models under simulated patient-specific consent restrictions. The comparison concerns permitted information use at inference time. It does not treat absence of consent as permission to collect data, and it does not evaluate removal of a patient's influence from an already trained model.

TRACER motivates the broader problem by combining continuous bedside video with clinical records for action recognition and longitudinal reasoning (Durante et al., 2026). Its authors report institutional review approval and restricted handling of protected information, while acknowledging consent and staff-surveillance questions for deployment. Their recordings are not publicly available. Our sepsis experiment therefore addresses a related permission-versus-performance question on public clinical data; it neither reproduces TRACER's tasks nor establishes comparable results on its private benchmark.

We frame this work as a sepsis benchmark study: a reusable protocol for evaluating specified permission interventions and the operating consequences of those interventions. The benchmark contributes paired comparisons and documented failure cases rather than a new learning algorithm or a claim to measure ethics as a single score.

The central empirical comparison holds the trained predictor, test patients, outcome definitions and alert thresholds fixed. It changes the permitted inputs through partial opt-out, denial of individual feature groups, complete physiological-input denial, and withdrawal during monitoring. Paired patient-level resampling estimates changes in discrimination, missed incidents, review demand and benchmark utility. Holding the decision rule fixed separates this comparison from our earlier sensitivity analysis, which selected a different empirical threshold for each consent scenario.

The mathematical analysis defines paired performance changes and a bounded-loss preservation criterion with a complete proof. It separates a statistical claim of non-inferiority at a declared tolerance from an ethical judgment about which information patients should permit. The same evaluation also examines alert timing, finite review capacity and modeled oversight, connecting changes in permitted information to the decisions and review requests that follow. The comparison follows hourly patient trajectories, including a specified consent-withdrawal time, rather than comparing isolated static predictions.

Related work

The PhysioNet challenge was designed to make early sepsis prediction comparable across cohorts, clinical variables and evaluation procedures (Reyna et al., 2020). Its utility rewards prediction at different times relative to onset and

penalizes false alarms. That makes it more informative than a single threshold-free ranking metric, but it still evaluates a prescribed prediction representation. Our once-per-patient review count and strict first-review window are additional estimands, not replacements for the challenge's original definition.

Clinical time-series benchmarks have established the value of comparing linear and recurrent models within a common preprocessing and evaluation framework (Harutyunyan et al., 2019). Missingness is also informative: the fact that a measurement was ordered can predict outcomes, independently of the recorded value. GRU-D explicitly represents missingness and time intervals (Che et al., 2018). Our GRU is a simpler baseline using causal summaries and observation counts; it does not implement learned decay or reproduce GRU-D's architecture. This distinction matters when interpreting the modest model search as evidence about a policy family rather than a comprehensive search for the best sepsis predictor.

Discrimination, probability quality and clinical usefulness also differ. Precision–recall analysis is particularly informative when positives are rare (Saito and Rehmsmeier, 2015); calibration studies emphasize that apparently strong ranking does not ensure reliable risk probabilities (Van Calster et al., 2019). We therefore tune using validation average precision and report AUROC, average precision, Brier score and challenge utility. Class-weighted training can improve the ordering of cases while changing the scale of the predicted probabilities, so we do not interpret the raw weighted scores as calibrated clinical risks.

The external validation of the Epic Sepsis Model examined both predictive performance and the burden of alternative alerting strategies (Wong et al., 2021). The TREWS adoption study examined whether clinicians evaluated alerts and how their interactions related to treatment timing (Henry et al., 2022). These studies motivate our separation between an alert being emitted and a correct action being completed. They do not supply the active service-time distribution for our simulated reviewers; response delay includes workflow and availability as well as time spent on the assessment.

Conformal risk control bounds an expected loss under exchangeability and the required regularity of a nested loss family (Angelopoulos et al., 2024). Risk-controlling prediction sets and Learn then Test instead provide high-probability statements about candidate procedures, with selection justified by valid tests or simultaneous confidence statements (Bates et al., 2021; Angelopoulos et al., 2025). Pareto Testing addresses the efficient selection of configurations subject to multiple risk constraints (Laufer-Goldshtein et al., 2023). Our finite-grid construction is deliberately elementary: exact binomial intervals and a union bound provide a transparent reference method for a non-monotone first-alert loss.

The distinction between the guarantees is consequential. An expectation over calibration and a future patient is not the same object as a high-probability upper bound on the population risk of a fitted rule. Likewise, failing to pass a test is not evidence that every candidate is infeasible. We use lower bounds to support the latter statement and retain an undetermined outcome when neither direction is established. We make no claim that the underlying binomial, union-bound or importance-weight arguments are new.

Recent clinical selective-prediction work combines conformal prediction with cost-aware deferral and evaluates group-conditional and shifted settings (Kwon and Kim, 2026). Such work is closely related in motivation. Our loss is a missed incident among event-positive patients, and our capacity variable is the number of patient reviews. Prediction-set coverage, error on retained cases and timely first review are different random variables, so a guarantee for one does not transfer automatically to the others. Under hospital shift, weighted conformal methods further require a relationship between source and target distributions (Tibshirani et al., 2019); an estimated domain-classifier ratio alone does not verify that relationship.

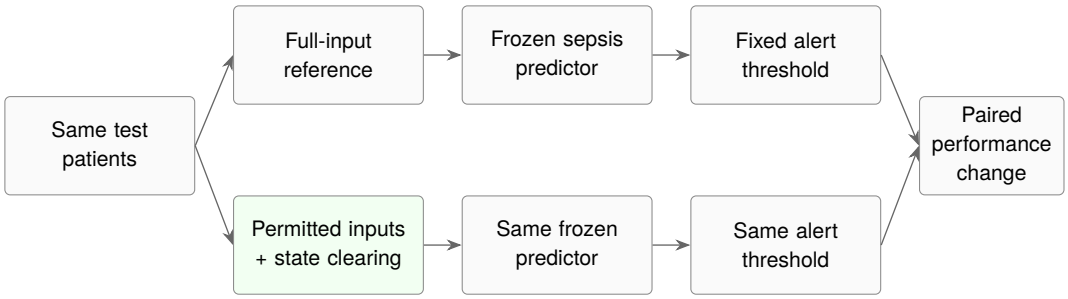
Dynamic consent treats permission as an ongoing interaction in which participants can manage preferences as data use evolves (Kaye et al., 2014). We model one technical consequence of changing permissions: which inputs and derived state are available to a predictor. This supports controlled sensitivity experiments but omits comprehension, communication, preference formation and the institutional process through which consent is obtained.

Meaningful human control has also been formulated in terms of systems responding to relevant human reasons and outcomes remaining attributable to responsible people (Santoni de Sio and van den Hoven, 2018). A review node in an information graph captures only a routing requirement. It says neither that a reviewer has enough information nor that the reviewer has time, competence or authority to act. Our graph checks and stochastic reviewer model address distinct parts of that larger process, while prospective human evaluation remains outside the present study.

Study design and data

The benchmark compares full-input AI with consent-constrained AI on the same held-out patients. We define the constrained system by the inputs it may process and the derived state it must clear. The experiment is designed to measure how that intervention changes performance, rather than assume that consent has either no cost or an unavoidable fixed cost. Sepsis provides a concrete, public evaluation setting; generalization to other diseases is not established by this study.

We use version 1.0.0 of the public PhysioNet/CinC 2019 release (Reyna et al., 2019, 2020). Hospital A contributes 20,336 patient files and 790,215 hourly records; hospital B contributes 20,000 files and 761,995 records. Together they contain 1,552,210 rows. Each record contains 40 input variables: eight vital signs, 26 laboratory variables and six demographic or administrative variables, plus the supplied



The consent comparison changes inference-time permissions while holding patients, labels, fitted model parameters and alert thresholds fixed. The restricted branch clears denied features and, after revocation, recurrent state. Paired effects measure the behavior of these fixed predictors; they are not effects of withholding treatment or retraining on a different cohort.

Public cohort and incident eligibility. "Positive" means at least one positive label. The two exclusion categories concern the strict incident endpoint; all patients remain in workload calculations.

Quantity	Hospital A	Hospital B
Patients	20,336	20,000
Hourly records	790,215	761,995
Label-positive patients	1,790	1,142
Initial-positive patients	203	223
Onset at/after observed end	2	13
Eligible incident patients	1,585	906

SepsisLabel. No video, skeleton or credentialed dataset contributes to the revised primary analysis.

The cohort is defined by the full public-file manifest, not by selecting a favorable subset. Downloads are checked against the published object ETag, and a SHA256 digest is recorded for each local file. Patient IDs, site membership, row boundaries and split assignments are retained in the released provenance. The public source is already de-identified; the study does not link records to external identities or collect new patient information. Dataset terms and evaluator licensing accompany the release.

The supplied label becomes positive six hours before inferred sepsis onset. Let a_i be the index of patient i's first positive label and set $o_i = a_i + 6$. We use zero-based hourly indices. The strict warning window is $[a_i, o_i)$, containing the six observations immediately before inferred onset. The label is not shifted a second time.

A patient enters the conditional incident-risk denominator when $a_i > 0$ and $o_i < T_i$, where T_i is the number of observed rows. Initial-positive patients have no observed pre-window prefix, and patients whose inferred onset is at or beyond the observed end do not meet this complete-window definition. The cohort table reports these exclusions. This retrospective convention is stricter than simply asking whether a positive label ever occurs.

For workload, positive patients are monitored through $\min(o_i, T_i)$ and patients without a positive label through T_i. All patients, including those excluded from incident risk, remain in the workload denominator. The stopping time depends on the retrospective label, which would not be known to a deployed model. It therefore defines a benchmark accounting interval, not a prospective operational stopping mechanism. Full-stay challenge utility is computed separately. A later deployment study would need an observable endpoint, such as discharge or an independently recorded clinical decision, and would have to reevaluate workload under that endpoint.

For each seed in $\{11, 29, 47\}$, a seeded permutation of hospital A assigns 55% to training, 15% to validation, 15% to calibration and 15% to testing. Integer cut points produce 11,184 training, 3,051 validation, 3,050 calibration and 3,051 test patients. Hospital B is independently permuted into 5,000 local-calibration and 15,000 external-test patients. No B labels train the clinical predictor.

Every observation from a patient stays in one partition within a seed. Hyperparameters and candidate thresholds use training and validation only. Calibration is reserved for risk/workload certification; test data evaluate the frozen result. The three seed-specific test sets overlap, so the analysis reports seed variability descriptively rather than treating the runs as three independent clinical cohorts. Splitting is not stratified on the outcome; the positive calibration counts are reported explicitly.

At hour t, each of the 40 raw variables contributes its most recent observed value up to t. For the 34 physiological variables, we additionally calculate the expanding mean of observed values and the cumulative observation count. The feature vector therefore has 40+34+34 = 108 entries. If x_{itv} is variable v and m_{itv} indicates observation, the summaries

are

$$c_{itv} = \sum_{u \le t} m_{iuv}, \qquad \bar{x}_{itv} = \frac{\sum_{u \le t} m_{iuv} x_{iuv}}{c_{itv}},$$
$$x^{\text{last}}_{itv} = x_{iu^*v}, \quad u^* = \max\{u \le t : m_{iuv} = 1\}.$$

The mean and last value remain missing until an observation exists; the count is initially zero. SepsisLabel is excluded from the input. Forward filling and expanding summaries use only the current prefix. The code does not interpolate backward from a later measurement. Tests perturb future observations and verify that earlier feature values do not change.

Observation counts intentionally retain information about the measurement process. This can improve prediction but also makes consent interventions consequential: masking a value while retaining its historical count would leave a prohibited information path. Missingness-aware clinical models motivate treating the observation process explicitly (Che et al., 2018); our count representation remains simpler than a dedicated missingness-decay architecture.

Prediction models and training

All models predict the hourly supplied label using the same causal representation. Training loss and the primary discrimination evaluation omit rows from inferred onset onward. For a positive patient, the final included row is $\min(o_i, T_i) - 1$; all observed rows of a label-negative patient are eligible. This prevents long post-onset stays from dominating the primary comparison. It does not remove the six pre-onset positive labels that define the prediction target.

The linear baseline imputes missing features with training-set means, retains initially empty feature columns, and applies training-set standardization. It fits an L2-regularized logistic model

$$s_{it} = \sigma(\beta^\top z_{it} + b),$$

where z_{it} is the transformed feature vector. Four candidates combine inverse regularization $C \in \{0.1, 1\}$ with either unweighted or class-balanced loss. The iteration limit is 350 and the random state follows the split seed. The selected model maximizes hourly validation average precision. This is a tuned linear baseline; it is not the fixed-C predictor used in the archived initial study.

The nonlinear tabular baseline uses histogram gradient boosting, building an additive tree score under binary classification loss (Friedman, 2001). Training-only variance screening removes constant and entirely missing columns; remaining missing values are handled by the estimator. The screening is necessary for the recorded implementation and uses no validation, calibration or test information.

The three candidates are $(15, 1, \text{unweighted})$, $(31, 5, \text{unweighted})$ and $(31, 5, \text{balanced})$, where the entries are maximum leaf count, L2 regularization and class weighting. Every candidate uses 120 boosting iterations, learning rate 0.08 and disabled automatic early stopping. Validation average precision chooses the candidate. The implementation uses scikit-learn (Pedregosa et al., 2011); the locked software version and exact parameter dictionaries are released. This small grid tests a stronger nonlinear predictor without claiming exhaustive hyperparameter optimization.

The sequence baseline has one unidirectional GRU layer (Cho et al., 2014) and a linear logit head at each hour:

$$h_{it} = \text{GRU}(z_{it}, h_{i,t-1}), \qquad s_{it} = \sigma(w^\top h_{it} + b).$$

Patient sequences are padded within batches and packed by their true lengths. Padding contributes no loss. The recurrent state is reset between patients. Although complete observed sequences can be passed through the forward computation, post-onset training positions are masked out of the loss; unidirectionality prevents later observations from influencing earlier outputs.

Training-set means and standard deviations normalize the 108 features. Empty or near-constant scales are replaced by one; normalized missing entries become zero and all normalized values are clipped to $[-10, 10]$. Candidate hidden sizes are 32 and 64. Each candidate receives six epochs with Adam (Kingma and Ba, 2015), learning rate 0.001, batch size 64 and gradient-norm clipping at one. Training patients are reshuffled each epoch using the split seed.

The positive-weighted binary cross-entropy is

$$\mathcal{L} = -\frac{1}{|\mathcal{I}|} \sum_{(i,t) \in \mathcal{I}} \left[\omega y_{it} \log s_{it} + (1 - y_{it}) \log(1 - s_{it})\right],$$

where $\mathcal{I}$ contains valid pre-onset training positions and $\omega = \min\{20, (1 - \bar{y})/\bar{y}\}$ uses the training prevalence. The implementation averages this loss over valid positions in each batch. Every epoch is evaluated on validation average precision; the best hidden-size/epoch pair is saved. There is no calibration-set early stopping, test-set model selection or bidirectional recurrence.

For all three model families, validation average precision is the sole model-selection criterion. AUROC is logged for the tabular candidates as a diagnostic. The selected estimators are frozen before calibration and before any consent or queue stress test. Prediction arrays preserve patient order and row boundaries for validation, A calibration, B calibration and both test sites. This allows later policy audits to reuse scores without retraining.

The search budgets differ by family: four logistic fits, three boosting fits and two six-epoch GRU trajectories per seed. They are appropriate for comparing these implementations, but they do not establish an architecture-level ranking under equal compute or equal tuning effort. We report both the settings and the selection rule so that a stronger model search can extend the study without changing its endpoints.

Matched consent comparison

The primary consent analysis reuses the nine fitted predictors and their original test splits. Model parameters, training data, labels, patient membership and validation-selected decision thresholds remain unchanged. The public full-input condition is the reference; it is not a claim that clinical data may be used without permission. The intervention applies only to inference-time availability. Consent to training, removal of training records, machine unlearning, actual preference collection and security against an attacker are outside this experiment.

There are eight prespecified conditions: full input; independent laboratory and vital-sign denial with probability 0.1, 0.3 or 0.5 per group; no laboratory inputs; no vital inputs; metadata only; and denial with probability 0.3 per group beginning at zero-based hour 12. Metadata remain available in every condition. Each patient's two random permission draws are fixed across models and rates, so higher denial rates form nested restrictions for that patient. The draws are keyed by patient, site and split seed and do not use outcomes. Probabilities concern each feature group; the probability of at least one denied group is $1 - (1 - p)^2$ at common rate p.

A denied value and its running mean become missing, while its observation-count features become zero. On withdrawal, denied historical features are cleared at the specified hour. A GRU restarts its recurrent state at that boundary, preventing its prior mixed state from carrying the denied information forward. This conservative reset also discards permitted historical context. Model imputation and normalization retain the transformations learned during the original training; a restricted model is not retrained or silently recalibrated.

For both first-alert and repeated-hourly policies, the threshold maximizes challenge utility on the original validation population and is then reused in every consent condition. Thus a paired effect includes the consequences of restricted inputs and the specified state reset under a fixed policy. It does not include a compensatory threshold change. The earlier scenario-adaptive consent analysis remains a separate secondary analysis and is not pooled with these paired effects.

Discrimination uses the same pre-onset rows in both conditions. We report AUROC, average precision and Brier score. Two hundred paired whole-patient bootstrap replicates estimate pointwise percentile intervals for differences in AUROC and Brier score; average precision is reported as a point estimate. Five hundred paired patient resamples estimate differences in conditional incident missed risk, reviews per patient, normalized challenge utility and false reviews per benchmark-negative patient. Each replicate uses identical patient multiplicities in the reference and restricted conditions, preserving the covariance of the comparison. Conditional missed risk recomputes its positive-patient denominator within each resample. Intervals condition on the trained model, test split and simulated permission assignments; they do not include retraining or real preference uncertainty.

We additionally count permitted physiological source-hours and the fraction of originally available last-value feature cells that remain available. These are information-availability proxies, not estimates of leaked information, re-identification probability or privacy protection. All scenarios retain the same outcome labels for evaluation. The experiment does not suppress treatment or simulate a change in disease caused by withholding data from the model.

The protocol specifies no clinically justified non-inferiority margin. Consequently, an interval containing zero is not interpreted as proof of equal performance, and an arbitrarily chosen tolerance is not presented as an ethical limit. Results are reported as signed changes with uncertainty for each model, hospital and split. Across-seed means and ranges are descriptive because the patient populations overlap.

We evaluate both a repeated-hourly policy and a first-only policy. At threshold τ, the repeated policy emits a review at every hour with $s_{it} \geq \tau$. Its missed-incident loss is one if no such hour lies in the six-hour warning window. The first-only policy emits at the first crossing and succeeds only if that crossing lies in the window; an earlier crossing consumes the review. Reviews are counted over the retrospective monitoring interval specified above. The repeated policy can exceed one review per patient, whereas the first-only policy cannot. These operational definitions prevent an improvement in ranking from being mistaken for unchanged incident detection.

The normalized challenge utility is

$$U = \frac{U_{\text{observed}} - U_{\text{inaction}}}{U_{\text{optimal}} - U_{\text{inaction}}},$$

using the official evaluator (Reyna et al., 2020). Repeated policies use the hourly threshold state. First-only policies latch a positive prediction state after their initial crossing; that persistent prediction is not counted as an additional review. Utility covers the full observed stay and uses the challenge's own timing rewards, so it differs from the strict incident-window and workload endpoints. False reviews per negative patient use only patients whose entire label sequence is negative.

AUROC measures ordering of positive and negative pre-onset hours; average precision summarizes the precision–recall curve; Brier score averages squared probability error. Whole-patient resampling preserves within-patient temporal dependence (Efron, 1979). Because of class weighting and rare positive hours, a lower Brier score need not imply improved event detection or reliable calibration. The benchmark therefore releases all these endpoints and does not collapse them into one score.

The full-input reproduction check verifies every saved predictor against its original prediction array before any mask is applied. The resulting reference-versus-reference differences and their bootstrap intervals are exactly zero. Released patient-level policy contributions, model hashes,

split assignments and permission configuration make the paired analysis independently inspectable. The benchmark protocol also separates future training-adapted or policy-adapted entries from these frozen full-input baselines.

Paired performance under consent restrictions

Let f be a predictor trained independently of the evaluation patients, and let M_i encode patient i's permitted inputs. Write $S_i^{\text{full}} = f(X_i)$ and $S_i^M = f(\Phi(X_i, M_i))$, where Φ includes causal masking and any required state clearing. For a fixed decision policy τ and bounded patient loss $L \in [0, 1]$, define

$$D_i = L(S_i^M, Y_i; \tau) - L(S_i^{\text{full}}, Y_i; \tau), \qquad \Delta = \mathbb{E}[D_i].$$

A positive Δ is deterioration under the specified restriction. A negative value is improvement for that particular loss. Neither sign measures the ethical value of respecting a patient's choice. Pairing compares both procedures on the same patient trajectory; it is a computational intervention on model access, not a causal estimate of changing patient health outcomes.

Proposition (Paired bounded-loss preservation criterion)**.** *For J fixed model/policy/permission comparisons on n independent, identically distributed evaluation units, let $\bar{D}_j$ be the observed mean loss difference. With probability at least $1 - \delta$, simultaneously for all comparisons,*

$$\Delta_j \le \bar{D}_j + \sqrt{\frac{2\log(J/\delta)}{n}}.$$

Thus an upper bound no larger than a prespecified tolerance ϵ certifies non-inferiority for that bounded loss and tolerance. It does not establish equality, clinical acceptability or ethical justification. For conditional incident risk, the units are incident-eligible patients and n is their count, conditional on membership.

Proof. Each paired difference lies in $[-1, 1]$. Hoeffding's one-sided inequality gives $\Pr(\Delta_j - \bar{D}_j > t) \le \exp(-nt^2/2)$. Set $t = \sqrt{2\log(J/\delta)/n}$ and union-bound the J events. Comparisons may share patients; independence is required between evaluation units, not between comparisons. The conditional argument applies to the positive-patient sampling law under the same fixed-procedure assumptions. □

This is a direct application of a classical concentration inequality (Hoeffding, 1963), not a new distribution-free result. AUROC is a ranking functional rather than a mean of these patient losses, so its paired bootstrap interval is reported separately and is not asserted to follow from this proposition. Repeated-review counts and normalized challenge utility also require their own range analysis before this bounded-loss statement can be applied.

Alert policies and joint guarantees

A patient supplies a finite sequence of causally available observations, consent states, and frozen scores $S_1, \ldots, S_T \in [0, 1]$. Let Z indicate an incident event with an observed valid warning window $[a, b]$. For a threshold τ, the first-alert policy is

$$T_\tau = \inf\{t : S_t \ge \tau\}, \quad R_\tau = \mathbf{1}\{T_\tau \le T\},$$
$$L_\tau = \mathbf{1}\{T_\tau \notin [a, b]\}, \quad Z = 1.$$

An empty set gives $T_\tau = \infty$. Assume $0 < \Pr(Z = 1) < 1$ and $0 < \alpha < 1$. We target conditional incident risk $r(\tau) = \mathbb{E}[L_\tau \mid Z = 1]$ and expected workload $w(\tau) = \mathbb{E}[R_\tau]$, measured in reviews per patient. Their joint constraints are $r(\tau) \le \alpha$ and $w(\tau) \le K$. For m independent patient arrivals in a period, the expected number of reviews is $mw(\tau)$; an expectation bound does not guarantee a finite waiting time.

For comparison, a repeated-alert policy uses every threshold crossing. Its incident missed-event loss is $\mathbf{1}\{\max_{a \le t \le b} S_t < \tau\}$, which is monotone in τ. A rule that emits only the first crossing has lower review count but a different risk.

Proposition (First-alert non-monotonicity)**.** *Nested threshold crossings do not imply monotone timely first-alert loss. There is a trajectory for which lowering τ changes the incident loss from zero to one, although the probability of any review is non-decreasing as the threshold decreases.*

Proof. Let a pre-window score equal 0.6, an in-window score equal 0.9, and all other scores equal zero. At threshold 0.8, the first alert lies inside the window and loss is zero. At threshold 0.5, the first alert occurs before the window and consumes the only alert; loss is one. Both thresholds trigger exactly one review. A threshold above 0.9 yields no alert and loss one. Thus the loss sequence as thresholds increase is $1, 0, 1$. Any-review events remain nested because they depend only on the overall score maximum. □

This prevents applying the monotone-loss conformal risk-control theorem (Angelopoulos et al., 2024) to the first-alert loss without additional assumptions. Counting any in-window crossing while reporting only one early review would evaluate a different policy.

Define the pre-window maximum $A = \max_{t<a} S_t$, with $A = -\infty$ for an empty prefix, the in-window maximum $B = \max_{a \le t \le b} S_t$, and the maximum C over the monitoring interval used to count reviews. For event-positive patients the monitoring interval includes the warning window. Let $\pi = \mathbb{P}(Z = 1)$ and

$$H(\tau) = \mathbb{P}(A < \tau \le B \mid Z = 1),$$
$$W(\tau) = \pi\mathbb{P}(C \ge \tau \mid Z = 1) + (1 - \pi)\mathbb{P}(C \ge \tau \mid Z = 0).$$

Theorem (Exact feasibility of a first-alert threshold family)**.** *For a fixed threshold domain $\mathcal{T}$, a jointly feasible first-alert*

threshold exists if and only if

$$\{\tau \in \mathcal{T} : H(\tau) \geq 1-\alpha,\ W(\tau) \leq K\} \neq \varnothing.$$

If $\mathcal{T}_\alpha = \{\tau \in \mathcal{T} : H(\tau) \geq 1-\alpha\}$ is nonempty and has a maximum $\tau_\star$, the minimum feasible expected review budget is

$$K_{\min} = W(\tau_\star).$$

Consequently, joint feasibility is equivalent to $K \geq K_{\min}$. Moreover $K_{\min} \geq \pi(1-\alpha)$. If $\mathcal{T}_\alpha$ is empty, no review budget makes this threshold family feasible.

The maximum assumption always holds for a finite nonempty policy grid. For a continuous domain it is a substantive attainment condition; replacing a minimum by an infimum without attainment can make a boundary feasibility assertion false. This characterization is exact but needs the joint law of A and B, not merely an hourly ROC curve.

Corollary (Closed form when early consumption is absent)**.** *Suppose no alert is possible before the valid window, the positive-patient review interval equals that window, and the conditional distributions of C are continuous, with positive-class CDF F_1 strictly increasing at its unique α-quantile q_α. Let the negative-class CDF be F_0. Then*

$$K_{\min} = \pi(1-\alpha) + (1-\pi)\{1 - F_0(q_\alpha)\},$$
$$q_\alpha = F_1^{-1}(\alpha).$$

This is the familiar threshold/ROC workload identity under an explicit no-early-consumption assumption; it is not a general formula for an unrestricted first-alert policy.

Let $\mathcal{T} = \{\tau_1, \ldots, \tau_M\}$ be fixed independently of calibration data, for example using training and validation only. Calibration patients are independent and identically distributed from the deployment population, including the consent mechanism. For each policy compute positive-patient losses and all-patient review indicators. Let n_+ be the positive calibration count and n the total count. Use exact one-sided Clopper–Pearson binomial bounds (Clopper and Pearson, 1934) with tail probability $\eta = \delta/(4M)$ to obtain (ℓ_j^r, u_j^r) and (ℓ_j^w, u_j^w).

Theorem (Joint certificate and honest abstention)**.** *Conditional on any independently trained predictor and frozen grid, with probability at least $1-\delta$ over calibration, all of the following hold simultaneously:*

Every j with $u_j^r \leq \alpha$ and $u_j^w \leq K$ satisfies both population constraints.

If every j has $\ell_j^r > \alpha$ or $\ell_j^w > K$, no policy on this grid satisfies both constraints.

Any data-dependent selection among certified policies, including the one with smallest workload upper bound, preserves both population constraints.

No monotonicity of L_τ is required. If neither certification nor grid exclusion is established, the output is undetermined, not infeasible. For $n_+ = 0$, the risk interval is $[0, 1]$ and no positive-risk claim is certified.

The construction applies simultaneous-confidence/LTT reasoning to the first-alert policy in the first-alert nonmonotonicity proposition. The three outputs distinguish an acceptable operating point, an excluded grid, and a calibration sample that cannot resolve the question.

The certificate is a high-probability statement about the population risk of the selected policy, distinct from the expectation-over-calibration guarantee of ordinary CRC. Model selection performed on calibration would require including every candidate model/policy in the multiplicity family; here model tuning uses validation only. Separate per-model certificates are not automatically simultaneous across models or seeds.

Corollary (Prespecified group or site cells)**.** *For groups $g = 1, \ldots, G$ specified independently of calibration, apply the joint-certification theorem within each cell using δ_g with $\sum_g \delta_g \leq \delta$. All certified group-conditional risk and workload statements then hold jointly with probability at least $1-\delta$. The groups may overlap. Empty or underpowered cells can remain undetermined. A rule calibrated in one hospital is not certified in another merely by relabeling its population.*

Let U denote the complete patient trajectory, C the possibly time-varying consent trajectory, and V any policy randomization. A frozen causal procedure $\Phi(U, C, V)$ produces permitted features, scores, alert decisions, and losses. The procedure may clear cached features and recurrent state after revocation.

Proposition (Sufficient exchangeability condition)**.** *If the calibration and future tuples (U, C, V) are identically distributed and independent, and the same frozen measurable transformation Φ is applied to every tuple, their loss and review vectors are identically distributed and independent. The joint-certification theorem therefore remains valid, including after conditioning on positive-event membership. Consent need not be independent of the outcome.*

The weaker condition $C \perp Y \mid X$ alone does not guarantee transfer between different consent populations. For a counterexample, let $X = Y \sim \text{Bernoulli}(1/2)$ and score $S = CX$. Calibration consent is always $C = 1$, deployment consent always $C = 0$, and the threshold is $1/2$. Both populations satisfy $C \perp Y \mid X$, yet positive-patient missed risk changes from zero to one. Conditional ignorability does not make the joint distributions equal.

Weighted conformal methods address particular distribution shifts when suitable likelihood-ratio information is available (Tibshirani et al., 2019). The known-ratio analysis gives a bounded-ratio risk/workload certificate derived from Hoeffding's inequality (Hoeffding, 1963). Its known-ratio assumption is stronger than fitting a hospital classifier. Our empirical weighted comparison is consequently a diagnostic, not a guaranteed hospital-transfer procedure.

Known-ratio transfer bound

Theorem (Known-ratio shift certificate)**.** *Suppose the target patient law Q is absolutely continuous with respect to source law P, with known density ratio $0 \le w \le B_w$ and $\mathbb{E}_P w = 1$. For a fixed grid and n independent source patients define*

$$\widehat{a}_j = n^{-1} \sum_i w_i Z_i L_{ij}, \quad \widehat{b} = n^{-1} \sum_i w_i Z_i,$$
$$\widehat{c}_j = n^{-1} \sum_i w_i R_{ij}.$$

Set $\epsilon = B_w \sqrt{\log((2M+1)/\delta)/(2n)}$. With probability at least $1-\delta$, simultaneously for every j with $\widehat{b} > \epsilon$,

$$\mathbb{E}_Q[L_j \mid Z = 1] \le \min\left\{1, \frac{\widehat{a}_j + \epsilon}{\widehat{b} - \epsilon}\right\},$$
$$\mathbb{E}_Q[R_j] \le \min\{1, \widehat{c}_j + \epsilon\}.$$

Thus selecting a policy whose two upper bounds meet (α, K) controls both target quantities. If the denominator is nonpositive, the risk bound is uninformative.

This is a bounded-importance-weight Hoeffding argument. A feature-only density ratio is valid only under the corresponding conditional outcome invariance and support assumptions. A fitted, clipped domain classifier does not provide a known ratio or a verified bound on approximation error. Our empirical weighted comparison is explicitly exploratory, not an application of the known-ratio theorem with a guaranteed ratio.

Proof of population feasibility

Proof of the population-feasibility theorem. A first alert is in the warning window exactly when no earlier score reaches threshold and some in-window score does. With inclusive alert ties, those events are $A < \tau$ and $B \ge \tau$. Thus $r(\tau) = 1 - H(\tau)$. Any review occurs exactly when $C \ge \tau$, giving $w(\tau) = W(\tau)$ by conditioning on Z. This proves the set characterization. The function W is nonincreasing in τ, regardless of whether H is monotone. If the risk-feasible set has maximum $\tau_\star$, its workload is no larger than at every other risk-feasible threshold, proving the minimum-budget identity and both directions of the budget equivalence. Timely success implies a review, so for every risk-feasible threshold $W(\tau) \ge \mathbb{P}(Z = 1, T_\tau \in [a, b]) = \pi H(\tau) \ge \pi(1-\alpha)$. If the risk-feasible set is empty, increasing workload cannot create a member of it. □

Proof of the no-early-consumption corollary. With no possible earlier alert and $B = C$ on positives, $r(\tau) = \mathbb{P}(C < \tau \mid Z = 1) = F_1(\tau)$ by continuity. The unique maximal feasible threshold is q_α under the stated strictness and attainment assumptions. Substitution into the workload decomposition gives $\pi\{1-F_1(q_\alpha)\} + (1-\pi)\{1-F_0(q_\alpha)\}$, and $F_1(q_\alpha) = \alpha$ completes the result. □

Proof of finite-sample certification

Proof of the joint-certification theorem. Condition on the fitted predictor and independently chosen candidate grid. For each policy, positive-patient losses are Bernoulli with mean r_j, conditional on the calibration positive count and membership; all-patient review indicators are Bernoulli with mean w_j. Inverting the binomial tails gives each one-sided interval coverage at least $1 - \eta$. This remains true unconditionally by averaging over the random positive count. Dependence between policies, or between risk and workload statistics, does not affect a union bound. There are $4M$ one-sided bounds, so all cover simultaneously with probability at least $1 - 4M\eta = 1 - \delta$. On this event, upper-bound feasibility implies true feasibility for every accepted policy, hence for any selection among them. If every policy has a lower bound violating at least one constraint, each policy violates that population constraint; therefore none on the grid is feasible. If neither condition is satisfied, the intervals do not resolve the question. The $[0, 1]$ convention at zero positives makes no unsupported claim. □

Proof of the group corollary. Apply the preceding coverage event within each prespecified group with error probability δ_g. A union bound over groups gives failure probability at most $\sum_g \delta_g$. Groups need not be independent. The conditional sampling assumptions must hold within each group, and post-hoc group discovery is not covered. □

Proofs under consent and shift

Proof of the consent-exchangeability proposition. For measurable sets $B_1, \ldots, B_n$, independence of the input tuples gives $\mathbb{P}(\Phi(U_i, C_i, V_i) \in B_i,\ i = 1, \ldots, n) = \prod_i \mathbb{P}((U_i, C_i, V_i) \in \Phi^{-1}(B_i))$. Identical input laws and the same transformation give identical output laws. Conditioning independent patient tuples on their own positive-event indicators preserves the corresponding conditional product law. Nothing in the argument requires consent to be outcome-independent; it requires the complete joint patient/consent mechanism to be stable across calibration and future data. □

Proof of the known-ratio theorem. The random variables wZL_j, wZ, wR_j all lie in $[0, B_w]$. One-sided Hoeffding bounds with failure probability $\delta/(2M+1)$ control the M numerator upper deviations, one denominator lower deviation, and M workload upper deviations. A union bound yields their simultaneous coverage. Change of measure gives $\mathbb{E}_P[wZL_j] = \mathbb{E}_Q[ZL_j]$, $\mathbb{E}_P[wZ] = \mathbb{P}_Q(Z = 1)$, and $\mathbb{E}_P[wR_j] = \mathbb{E}_Q[R_j]$. When the denominator lower bound is positive, dividing its numerator upper bound by that lower bound bounds the target conditional risk. Truncation at one uses the Bernoulli range. The same simultaneous event covers data-dependent selection among accepted candidates. □

Consent, calibration and oversight evaluation

The joint policy analysis asks whether a frozen predictor and threshold policy can meet two declared constraints simultaneously: conditional incident missed-event risk at most $\alpha = 0.05$ and expected review workload at most K. We examine $K \in \{0.1, 0.25, 0.5, 0.75, 1\}$ reviews per patient. These values define a sensitivity analysis over review budgets; they are not estimates of capacity at a particular hospital. The statistical confidence parameter is $\delta = 0.05$ within each model and frozen policy grid.

We also evaluate benchmark usefulness using positive normalized challenge utility. This is an operational comparison with the challenge's inaction reference, not a clinical benefit threshold. The full research question additionally requires permitted data flow and effective review. Structural checks assess the specified consent and review graph, while the queue experiment measures timely and correct action only within a declared simulation. We do not combine these evidence types into a single clinical-safety score.

For each frozen predictor, we pool its validation scores and take 51 equally spaced empirical quantiles from 0 to 1, adding thresholds 0 and 1.000001. Duplicate values are removed. The first endpoint alerts whenever monitoring is nonempty, and the last emits no alert for scores in $[0, 1]$. The resulting grid is independent of the calibration and test outcomes conditional on training and validation.

For each calibration patient and threshold, the code records an incident-loss indicator, an any-review indicator and the time of the first crossing. Conditional risk uses only incident-eligible patients; workload uses every calibration patient. Let k_j^r be missed incidents among n_+ positives and k_j^w be reviewed patients among n total patients. For either binomial count (k, N), an exact one-sided interval with tail probability η has endpoints

$$\ell(k, N) = \begin{cases} 0, & k = 0, \\ \mathrm{Beta}^{-1}(\eta; k, N - k + 1), & k > 0, \end{cases}$$

$$u(k, N) = \begin{cases} 1, & k = N, \\ \mathrm{Beta}^{-1}(1 - \eta; k + 1, N - k), & k < N. \end{cases}$$

We set $\eta = \delta/(4M)$ for M grid policies, applying the bounds to both risk and workload (Clopper and Pearson, 1934). A zero-size sample receives $[0, 1]$. The joint-certification theorem then justifies the three outputs used throughout the results.

Among certified policies, the implementation selects the smallest workload upper bound. If none is certified, it does not deploy an empirically attractive policy under the name of a certificate. A grid-infeasible result requires every candidate to violate at least one lower-bound test; otherwise the result remains undetermined. Computation uses $O(nM)$ indicator work and storage in the present implementation. Streaming counts could reduce storage, but no such optimization is needed for these cohort sizes.

Joint certification on a frozen grid

Require: Patient losses L, reviews R, eligibility Z
Require: Frozen thresholds $\mathcal{T}$, targets (α, K, δ)
  $M \leftarrow |\mathcal{T}|$; $\eta \leftarrow \delta/(4M)$
  for $j = 1, \ldots, M$ **do**
    Compute exact risk bounds (ℓ_j^r, u_j^r) on $Z = 1$
    Compute review bounds (ℓ_j^w, u_j^w) on all patients
  end for
  $F \leftarrow \{j : u_j^r \le \alpha \text{ and } u_j^w \le K\}$
  if $F \neq \varnothing$ **then**
    return certified, $\arg\min_{j \in F} u_j^w$
  else if $\ell_j^r > \alpha$ or $\ell_j^w > K$ for every j **then**
    return grid infeasible
  else
    return undetermined
  end if

Finite calibration size is relevant even when no misses are observed. For $k = 0$, the upper risk bound is $1 - \eta^{1/n_+}$. Thus zero observed misses can meet α only when

$$n_+ \ge \left\lceil \frac{\log \eta}{\log(1 - \alpha)} \right\rceil.$$

For $M = 53$, $\alpha = \delta = 0.05$, this requires 163 positives. The actual local calibration pools contain 219–253 positives in A and 230–242 in B. Their failure is therefore not explained solely by a correction that would reject even a zero-miss rule.

We distinguish three evaluations: A calibration with A testing, A calibration with B testing, and B calibration with B testing. The first and third target a locally matched patient distribution. A frozen A rule has no asserted B guarantee merely because the same score function is used. B recalibration changes the threshold evidence, not the clinical predictor, which remains trained on A.

Four demographic cells are prespecified: female, male, age below 65 and age at least 65. Sex and age cells overlap. Each receives failure budget $\delta/4$ and is evaluated at $K = 0.5$. The code reports positive calibration support and group-specific certificate status. Sparse groups may remain unresolved even when a population-wide claim is available. The analysis does not infer demographic fairness from an aggregate certificate, and the available binary recorded gender variable does not represent the full range of gender identities.

A hospital-shift diagnostic fits a logistic domain classifier to the first available feature row of A training patients and B calibration patients. Its estimated B probability $q(x)$ gives a sample-size-adjusted density-ratio estimate, clipped to $[0.05, 20]$. On incident-positive A calibration patients, we compute weighted repeated-window loss and a CRC-inspired correction using the maximum fitted weight. The effective positive sample size is $(\sum_i w_i)^2 / \sum_i w_i^2$. The resulting threshold is tested on B. Because the ratio is estimated and clipped and conditional invariance is unverified, this diagnostic is not an implementation of the known-ratio

theorem with an exact guarantee.

To examine the role of calibration size, we repeatedly sample 25, 50, 100, 200 or 400 incident-positive calibration patients without replacement, when the local pool is large enough. There are 200 draws per available size, model, site and seed. Within a draw, a simultaneous one-sided risk upper bound uses tail probability $\delta/(2M)$. If multiple policies pass, the policy with the lowest empirical review frequency in the all-patient calibration pool is selected. This analysis certifies risk only; it does not add a workload guarantee to the subset experiment.

We record whether a rule is certified, its realized test risk when it is, and whether that risk exceeds α. Draws reuse the same calibration pool and test set and are therefore dependent. Their empirical frequency of exceedance is a stability diagnostic rather than a binomial estimate from independent deployment trials. If no rule is certified, the conditional exceedance frequency is undefined. It must not be reported as zero or interpreted as evidence of perfect coverage.

Primary hourly discrimination uses only the pre-onset rows defined in the study design. AUROC measures ranking across positive and negative rows; average precision summarizes the precision–recall curve; Brier score is the mean squared difference between a score and the binary label. Average precision is particularly relevant to rare positive labels (Saito and Rehmsmeier, 2015). Brier score reflects overall probability accuracy and should not be labeled a pure calibration metric (Van Calster et al., 2019).

For every model, site and seed, we draw 500 bootstrap samples of whole patients with replacement. All rows of a sampled patient receive the same multiplicity. AUROC and Brier score are recomputed using those multiplicities, and the 2.5th and 97.5th percentiles form the reported intervals. This cluster-level use of the bootstrap (Efron, 1979) preserves within-patient temporal dependence and treats patients as the sampling units. It conditions on the fitted model and split; it does not include uncertainty from a new model search or a new hospital population.

The normalized challenge utility is

$$U = \frac{U_{\text{observed}} - U_{\text{inaction}}}{U_{\text{optimal}} - U_{\text{inaction}}},$$

using the official additive utility implementation (Reyna et al., 2020). For repeated alerts, predicted condition state is the hourly threshold decision. For the first-alert policy, the state remains positive after the first crossing. This latched state represents persistent positive prediction, not a new review at every later hour. Utility is evaluated over the full stay, while review count and incident loss follow the retrospective monitoring definition. Tests compare the vectorized utility calculation against the pinned official evaluator.

A separate operating-point audit selects the threshold that maximizes validation utility for each model and policy, then freezes it before evaluating either test site. For these fixed policies, exact pointwise 95% binomial intervals describe conditional incident risk. They are distinct from the simultaneous calibration certificate. The audit also decomposes first-alert misses into an early review and no review before onset. Test-frontier points are used only to describe the tradeoff surface, never to choose the reported operating policy.

We simulate permission for two input groups, laboratory measurements and vital signs. For each patient, each group is denied independently with probability $p \in \{0, 0.1, 0.3, 0.5\}$. Thus p is a per-group probability, not the proportion of patients with any denial; the latter is $1 - (1 - p)^2$. Demographic and administrative variables remain available. The experiment studies heterogeneity across patients rather than uniformly removing a feature block from the whole population.

In the admission opt-out scenario, denied last values and expanding means are set to missing throughout the trajectory, while denied observation counts are set to zero. This removes both recorded values and the associated measurement-history summaries. The previously fitted imputer or missing-value handling is then applied without retraining. The intervention does not infer values from future observations or preserve a denied historical count.

In the revocation scenario, the same denial decision takes effect at zero-based hour 12. Denied summaries are cleared from that point onward, including information accumulated before withdrawal. For the GRU, the sequence is split at withdrawal and the subsequent segment starts with a new hidden state. This conservative reset also loses permitted information encoded in the previous state; the experiment does not attempt selective deletion from a mixed latent representation.

For each scenario, calibration and both test sites are rescored using the frozen predictor. A calibration set from A is used to seek a joint certificate at $K = 0.5$. When no certificate exists, the descriptive cost analysis uses a labeled empirical fallback: minimum-workload threshold among those with empirical risk at most 0.05, or minimum empirical risk if none qualifies. This fallback has no asserted risk guarantee. Consequently, consent curves combine the score intervention with scenario-specific empirical threshold selection and are not a fixed-threshold causal effect.

Scenario seeds are reused to make permission patterns reproducible and comparisons stable. The stress experiment is not an independent repeated-sampling validation of the consent theorem. That theorem requires the complete patient/consent tuples to obey its stated sampling law. Real preferences, correlations between consent and health status, communication quality, and strategic selection are not observed in this dataset.

The queue uses the validation-selected first-alert threshold and the actual held-out patient score trajectories. Patient admission times are generated by exponentially distributed interarrival times with rates 0.5, 1 or 2 patients per hour. Each patient's request occurs at simulated admission plus that patient's first-alert hour. Requests are sorted by arrival time and handled by one first-come, first-served reviewer.

For request j at time a_j, constant service time s and pre-

vious completion d_{j-1}, waiting and completion are

$$q_j = \max\{0, d_{j-1} - a_j\}, \qquad d_j = \max\{a_j, d_{j-1}\} + s.$$

Service times are 2, 5 or 10 minutes. Base reviewer sensitivity is 0.8, 0.9 or 1, and specificity is 0.9 or 0.99. A declared fatigue rule multiplies sensitivity by $\max\{0.5, 1 - 0.05q_j\}$ when waiting is measured in hours. The resulting sensitivity governs a Bernoulli correct-review draw. A false consequential action is drawn with probability one minus specificity only for patients whose entire label sequence is negative.

An incident succeeds end to end only if its first alert lies in the valid warning window, review is correct, and completion occurs before inferred onset. A perfect reviewer cannot recover an incident with no timely request under this policy. The simulation reports mean and 95th-percentile waiting, review count, false actions and conditional end-to-end missed risk. The $3 \times 3 \times 3 \times 2$ parameter combinations give 54 scenarios per model, site and seed, or 972 overall.

These service, arrival and fatigue values are sensitivity assumptions. The TREWS study supplies evidence that response and adoption must be measured (Henry et al., 2022), but response delay is not an estimate of active review duration. The queue therefore does not establish a staffing recommendation, a hospital-specific workload limit or an empirical model of clinician cognition. The corrected audit excludes all label-positive patients from the false-action class, including positives who are ineligible for the strict incident endpoint.

The structural component represents data sources, transformations, storage, review and consequential actions as a directed graph. Each edge carries the patient and processing purpose to which permission applies. Applying a consent mask removes prohibited flows; withdrawing a source invalidates cached state that depends on it. This model makes the tested policy explicit, but it does not inspect an external hospital application or prove that every real data flow has been represented.

Mandatory review is checked as a path property. For sensitive source set S, consequential destinations D and review nodes H, every S-to-D path intersects H exactly when deleting H disconnects S from D. This standard equivalence is tested by reachability after node removal. A reviewer placed somewhere in a graph is insufficient if a bypass remains.

To move beyond the original small constructed graph, we generate 500 DAGs with 50–300 nodes. For ordered nodes $i < j$, an edge is included independently with probability $3/n$. Approximately 10% of internal nodes are assigned as review nodes; node 0 is the source and node $n - 1$ the action. We report whether an action path exists and whether one survives removal of review nodes. These simulations assess structural behavior across a broader family. They do not estimate the prevalence of bypasses in deployed clinical systems, and they do not turn a path constraint into evidence of effective human review.

Results

The paired comparison evaluates eight input-permission conditions for three models, three split seeds and two hospitals. It produces 144 discrimination records and 288 fixed-policy records. Full-input scores are recomputed and checked against the saved parent predictions before any intervention. Every reference-versus-reference contrast has zero difference by construction; restricted comparisons use the same patients and thresholds rather than independently selected test samples.

Every non-reference permission condition reduces mean AUROC within each model–hospital combination, although the size of the change varies substantially. At hospital A, boosting changes from 0.794 with full inputs to 0.780, 0.763 and 0.747 at per-group denial probabilities of 0.1, 0.3 and 0.5. Its corresponding hospital-B values are 0.731, 0.716, 0.688 and 0.662. At probability 0.3, the mean AUROC decrease ranges from 0.019 to 0.051 across the six model–hospital combinations. These averages describe three overlapping patient splits; they are not independent-cohort meta-analytic estimates.

The source being restricted matters as much as the overall amount of available information. For the GRU at hospital B, removing all laboratories changes mean AUROC from 0.752 to 0.745, whereas removing all vitals reduces it to 0.657. Under laboratory restriction, the paired AUROC differences for seeds 11, 29 and 47 are respectively −0.0012, −0.0086 and −0.0107. Their pointwise 95% paired bootstrap intervals are $[-0.0113, 0.0096]$, $[-0.0226, 0.0027]$ and $[-0.0189, -0.0024]$. The first two intervals include zero, but that does not establish equivalence. Under vital restriction, all three corresponding intervals lie below zero. These comparisons motivate source-specific evaluation rather than a single claim that less data always causes a fixed performance penalty.

Ranking similarity can conceal an operating change that the ranking metric alone does not reveal. For the same GRU at hospital B, laboratory restriction increases mean conditional missed-incident risk from 0.534 to 0.646 at the fixed repeated-hourly threshold and reduces reviews from 4.172 to 2.756 per patient. The increase in missed risk is 10.1, 10.2 and 13.3 percentage points across the three seeds; all three pointwise paired intervals exclude zero. The small average AUROC loss therefore does not imply preservation of alert sensitivity. At hospital A, a 0.3 denial probability increases GRU missed risk by 15.8 percentage points while reducing reviews by 1.69 per patient.

The direction of workload change is model-dependent. At hospital A, removing laboratory inputs from boosting decreases missed risk by 4.5 percentage points but adds 2.51 reviews per patient and decreases normalized challenge utility by 0.034. With 0.3 denial probability, boosting adds 1.13 reviews per patient at A and 1.53 at B, while the GRU and logistic model request fewer reviews. These are fixed-policy responses to changed score trajectories, not consent-specific threshold optimizations. A reduction in review vol-

ume can accompany more misses, and fewer misses can be purchased by a larger review burden.

Probability scores also require separate interpretation. Some restrictions improve Brier score while worsening AUROC. Class-weighted full-input predictors are not probability-calibrated, and the benchmark has rare positive hours; a shift toward lower predicted probabilities can improve mean squared probability error without preserving event detection. No single metric is therefore treated as a sufficient measure of retained usefulness.

With no physiological inputs, mean AUROC ranges from 0.525 to 0.710 across model–hospital combinations. This condition retains the benchmark's demographic, administrative and time-related features, so it is not an information-free prediction task. Withdrawal at hour 12 also produces a distinct pattern: GRU mean AUROC falls to 0.717 at A and 0.698 at B. Its conservative state reset removes mixed historical information, including permitted context; the observed change consequently includes that implementation choice.

The permission-volume indicators are descriptions of the intervention. At a 0.3 denial probability, approximately 70% of physiological source-hours remain permitted. These counts do not measure privacy leakage, the number of unique clinical observations, or the value of particular information to a patient. They make the permission intervention inspectable without presenting a scalar privacy score.

These comparisons answer a narrower, explicit version of the consent question: what happens to a frozen sepsis AI system when its permitted inputs change? They do not establish that the observed losses are unavoidable. Retraining under permitted features, adapting missingness handling or choosing a new validation-only threshold could change the outcome. Those are different interventions that would require separately evaluated models, rather than being folded into the same consent contrast. The present results also do not identify a clinically acceptable tolerance, so apparent similarity on an individual metric is not labeled non-inferiority. The paired comparisons describe how input restrictions change a fixed system. We next ask whether the resulting prediction and review machinery can satisfy an explicit timing and capacity requirement, examining first-alert policies, calibration evidence and simulated human review within the same evaluation.

The discrimination table summarizes all three splits. Boosting has the highest mean AUROC in A; the GRU has the highest mean AUROC in B. These are descriptive rankings: the seed-specific test populations overlap. Average precision remains low, and the class-weighted models are not calibrated probabilities. The complete patient-bootstrap intervals and Brier scores are reported in the seed-specific uncertainty table.

For every model and seed, the local A and B calibrations certify grid infeasibility at $\alpha = 0.05$, for every tested budget from 0.1 to 1 review per patient. There are 90 local calibration/budget outcomes; each certificate retains its stated per-grid confidence interpretation. The result concerns the frozen grids, not all thresholds or all monitoring designs. Increasing review capacity cannot repair a grid whose first-alert timing already violates the risk target.

Pre-onset discrimination, mean across three seeds. AP denotes average precision. Per-seed 95% patient-bootstrap intervals appear in the seed-specific uncertainty table; seed averaging is not an independent-cohort meta-analysis.

Model	AUROC A	AUROC B	AP A	AP B
Logistic	0.745	0.666	0.049	0.025
Boosting	0.794	0.731	0.060	0.033
GRU	0.778	0.752	0.060	0.031

The risk–workload figure also shows the realized test risks. Even the best test-grid point, selected retrospectively only to describe the frontier, misses at least 90.0% of eligible incidents. This stringent endpoint treats a review just before the six-hour window as a miss, even if it might be useful clinically. The negative result is therefore about timely *first* review under this definition, not about whether early alerts have clinical value.

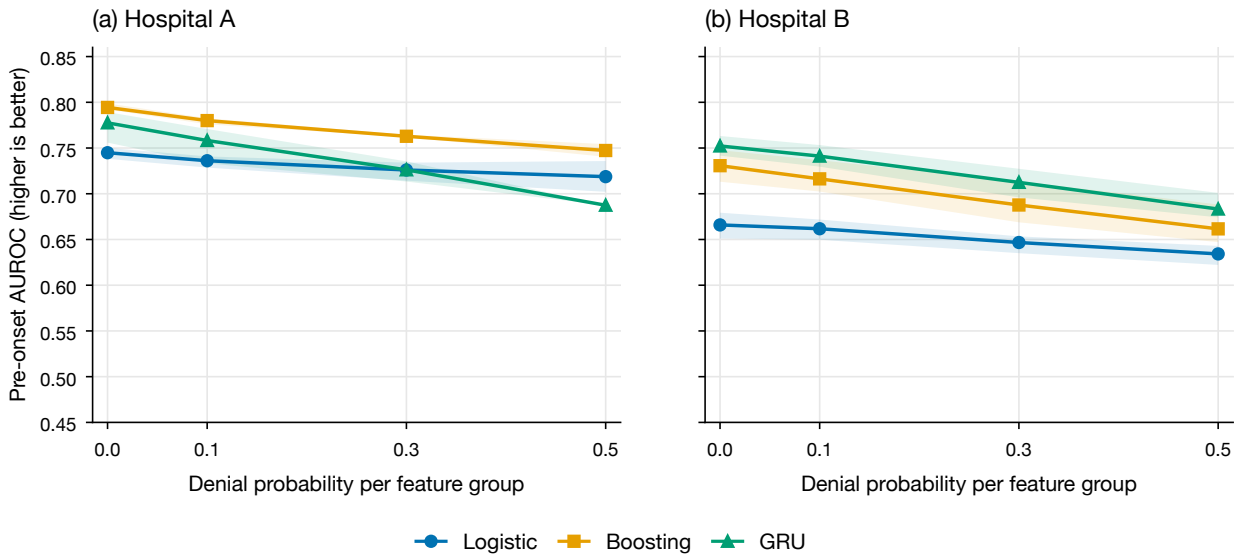
We additionally choose thresholds by validation challenge utility and freeze them before testing (the operating-policy table). Boosting in A achieves mean utility 0.332 with 0.193 reviews per patient, yet its strict incident missed risk is 0.960. Its three test splits contain 742 eligible incident observations in total, counting overlap across splits: 414 receive an early first alert and 298 receive none before onset. Only 30 receive their first alert in the specified window. The challenge utility credits a broader time window and a persistent positive state; it is not a surrogate for this first-review endpoint.

Repeated alerts recover more incidents but require several reviews per patient. At validation-selected utility thresholds, no model/site/policy combination has all three of positive utility, risk at most 0.05, and workload at most 0.5. This is the empirical answer to the declared joint target, not proof against alternative cooldown or continuing-review policies.

The first-alert outcomes figure separates the two mechanisms. An earlier review is not equivalent to an unreviewed patient, but both fail the strict first-review window. This decomposition should accompany any interpretation of the aggregate missed-risk value.

Local B calibration contains 230–242 eligible positives, compared with 219–253 in A. None of 14,400 overlapping calibration subsets (25, 50, 100 or 200 positives) certifies the 5% first-alert risk target. Thus the fraction of test-risk violations *among certified rules* is undefined, not zero. Four hundred positives are unavailable in these calibration pools; the study does not establish how much additional data would suffice. All 72 prespecified group/model/site/seed checks also exclude their tested grids.

Consent interventions do not restore feasibility. They change both scores and the descriptive fallback threshold, so a reduced review count need not be an improvement. For



Pre-onset discrimination under increasing independent denial of laboratory and vital inputs. Points are means over three matched split seeds and shading spans their observed range, not a confidence interval. The vertical scale is restricted to emphasize differences in AUROC. Models are frozen and no consent-specific retraining occurs.

Validation-selected operating policies, means over three seeds. "Repeated" reviews every above-threshold hour. Utility uses full-stay states; risk and review count use the retrospective pre-onset endpoint. No test metric chooses a threshold.

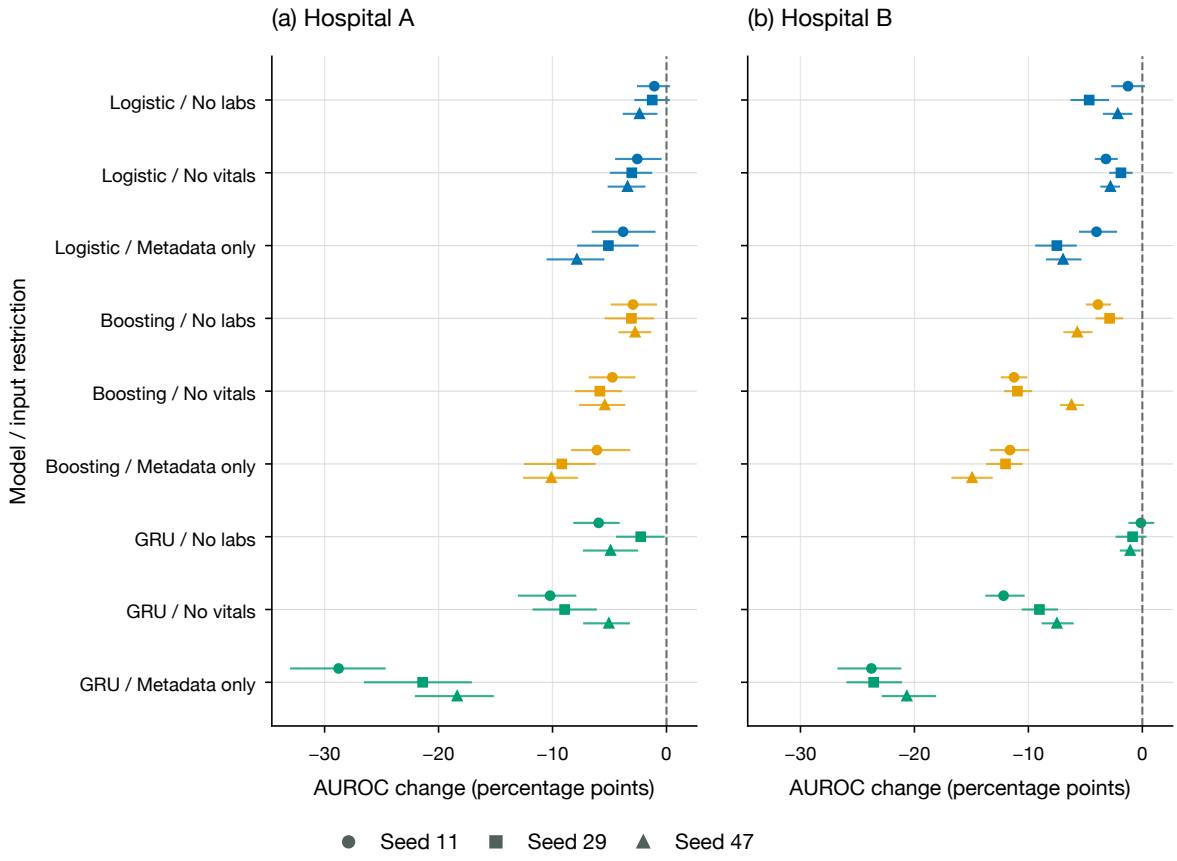
Model	Site	Policy	Missed risk	Reviews/patient	Utility
Logistic	A	First	0.936	0.215	0.333
Logistic	A	Repeated	0.298	8.077	0.342
Logistic	B	First	0.916	0.296	0.085
Logistic	B	Repeated	0.302	10.185	0.003
Boosting	A	First	0.960	0.193	0.332
Boosting	A	Repeated	0.318	6.741	0.380
Boosting	B	First	0.970	0.091	0.125
Boosting	B	Repeated	0.494	4.162	0.183
GRU	A	First	0.949	0.160	0.347
GRU	A	Repeated	0.406	6.124	0.352
GRU	B	First	0.971	0.081	0.155
GRU	B	Repeated	0.534	4.172	0.180

example, GRU opt-out at 50% per feature group changes mean A missed risk from 0.952 to 0.979 while reviews fall from 0.207 to 0.104 per patient. These are simulated permissions, not observed patient preferences. The estimated-ratio repeated-alert diagnostic has external-site risk from 0.034 to 0.152 across model/seed combinations and carries no exact shift guarantee.

The corrected reviewer audit evaluates 972 queue scenarios on frozen scores. Across all these scenarios, end-to-end missed risk ranges from 0.912 to 0.987; a perfect reviewer cannot recover an incident that never receives a timely review. The simulation is not evidence about real clinician performance. Among 500 random DAGs, 126 contain a source-to-action path and 103 contain a path bypassing the randomly assigned review nodes. This illustrates why review nodes alone do not enforce review, while remaining a synthetic structural test.

The consent-cost figure shows why fewer reviews cannot be interpreted in isolation. The GRU's review demand falls as more input groups are denied, while its already high missed risk increases. Logistic and boosting responses are less monotone because permission changes alter the score trajectories and the empirical fallback threshold. The released revocation analysis additionally separates admission opt-out from clearing state at hour 12. Neither intervention recovers a certified first-alert operating point. The complete seed-specific discrimination intervals and feasibility assessments retain the variation behind the across-seed summaries.

The policy audit reports 36 validation-selected operating points with exact pointwise 95% binomial intervals in `selected_policies.csv`. They are not simultaneous across models, sites and seeds. The confidence intervals in that table assess realized test risk; the certificate uses an



Paired AUROC differences for complete feature-group restrictions. Each point is one split seed and each interval is the pointwise 95% percentile interval from 200 paired whole-patient bootstrap replicates. The same patient multiplicities are used in the full and restricted conditions. Intervals are not simultaneous across the displayed comparisons. Zero denotes unchanged AUROC, not an ethical or clinical acceptability threshold.

All-seed pre-onset AUROC with 95% percentile intervals from 500 whole-patient bootstrap replicates. Brier scores are point estimates; their intervals are released in the machine-readable metrics table. Intervals describe patient-sampling uncertainty conditional on each fitted model.

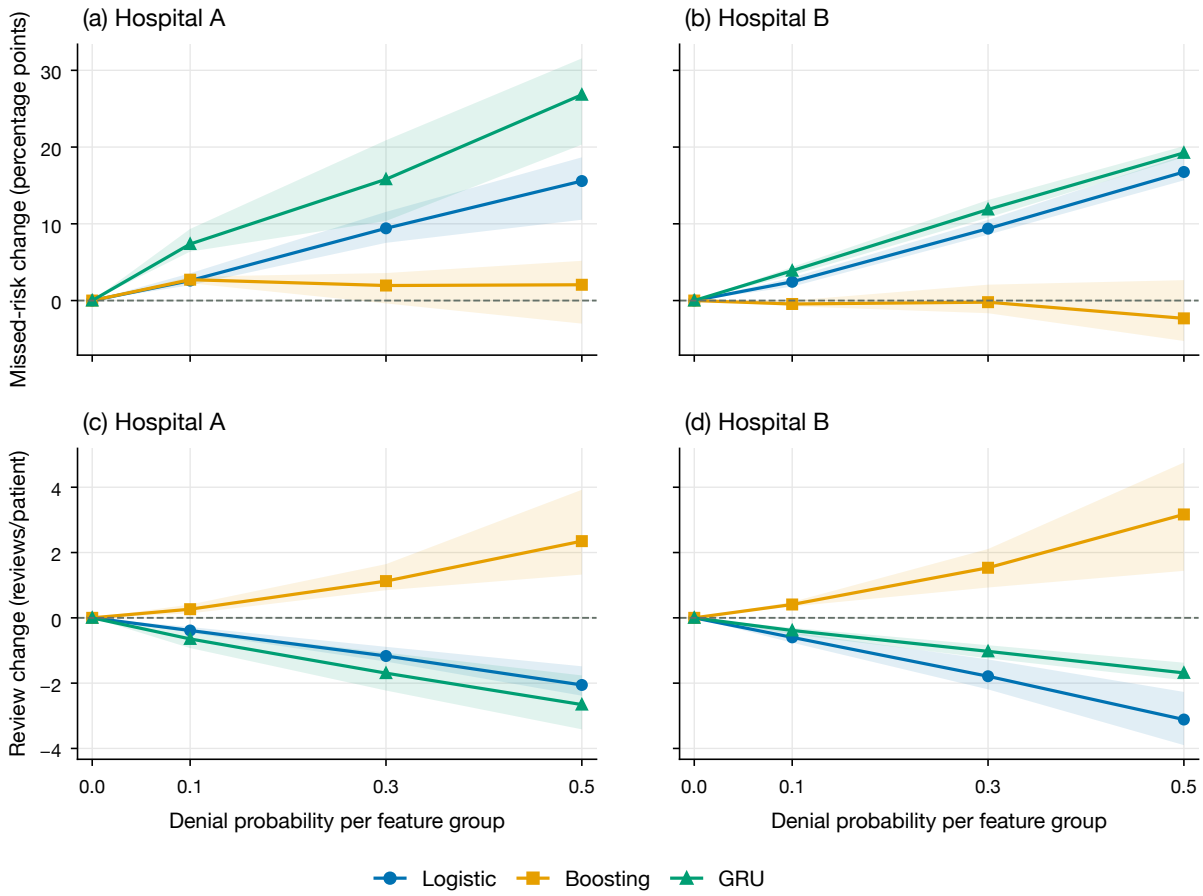
Model	Seed	AUROC A [95% CI]	AUROC B [95% CI]	Brier A	Brier B
Logistic	11	0.750 [0.724, 0.776]	0.651 [0.634, 0.669]	0.015	0.014
Logistic	29	0.738 [0.711, 0.766]	0.679 [0.661, 0.695]	0.187	0.235
Logistic	47	0.746 [0.718, 0.773]	0.667 [0.649, 0.682]	0.189	0.210
Boosting	11	0.792 [0.769, 0.818]	0.713 [0.696, 0.732]	0.098	0.060
Boosting	29	0.799 [0.775, 0.821]	0.731 [0.715, 0.746]	0.092	0.055
Boosting	47	0.792 [0.767, 0.816]	0.748 [0.730, 0.763]	0.014	0.009
GRU	11	0.789 [0.766, 0.816]	0.742 [0.726, 0.759]	0.056	0.043
GRU	29	0.787 [0.761, 0.813]	0.752 [0.737, 0.766]	0.057	0.036
GRU	47	0.756 [0.732, 0.782]	0.763 [0.749, 0.777]	0.066	0.053

independent calibration population. No certified policy is available for a certified-policy test-risk distribution.

The paired discrimination table reports every model–site combination across the eight input conditions, completing the comparison between full inputs and permission-restricted inputs.

Discussion

The matched experiment directly addresses whether a frozen sepsis AI system retains its behavior under consent-related input restrictions. The answer is conditional: the magnitude and operational meaning of the change depend on the model, hospital, restricted source and endpoint. All model–hospital mean AUROCs decline under the tested restrictions. Some declines are small, but a small ranking



Changes in missed-incident risk and review demand at fixed repeated-hourly alert thresholds. Positive risk change is deterioration; negative review change means fewer reviews. Lines are three-seed means and bands are descriptive seed ranges. All thresholds were selected on the original validation population and remain fixed across the denial-rate sweep. Paired policy confidence intervals are released separately.

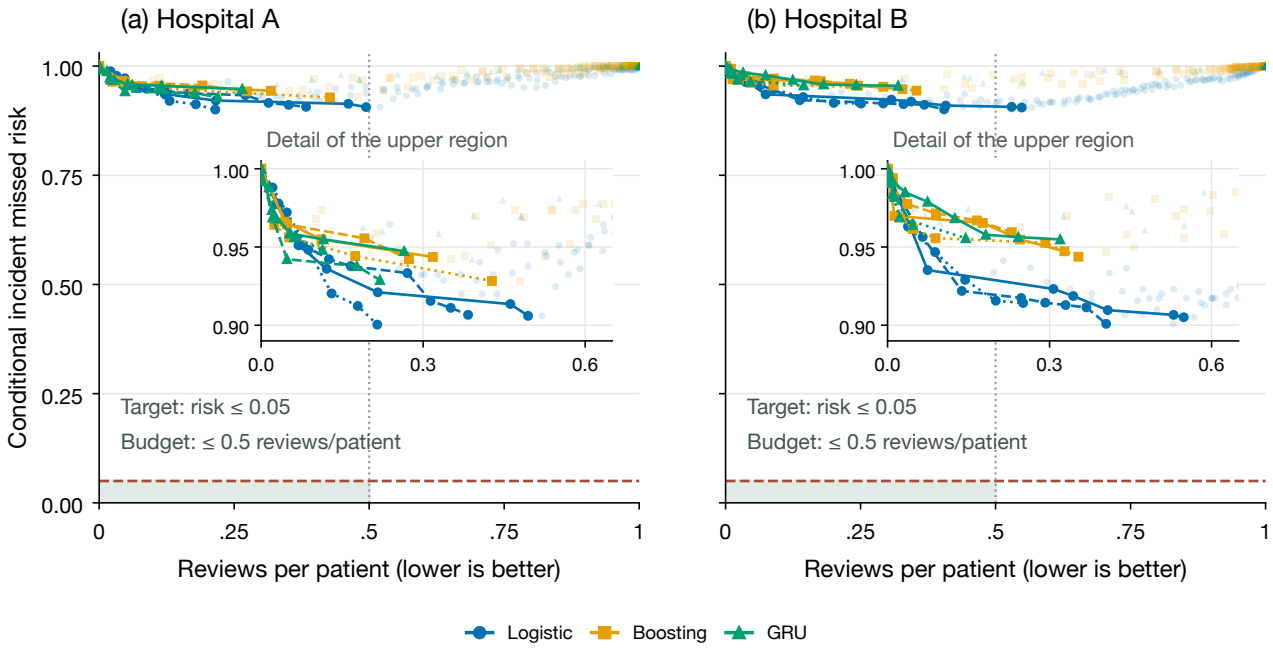
Feasibility of the validation-selected first-alert policy under conditional incident loss. Each row aggregates three seeds: "Y" means every seed meets the empirical criterion; "N" means none does. Storage and review checks concern the specified graph, not a deployed system. Actual human oversight was not measured.

Model	Site	Utility > 0	Risk ≤ .05	Reviews ≤ .5	Storage cut	Review path
Logistic	A	Y	N	Y	Modeled	Modeled
Logistic	B	Y	N	Y	Modeled	Modeled
Boosting	A	Y	N	Y	Modeled	Modeled
Boosting	B	Y	N	Y	Modeled	Modeled
GRU	A	Y	N	Y	Modeled	Modeled
GRU	B	Y	N	Y	Modeled	Modeled

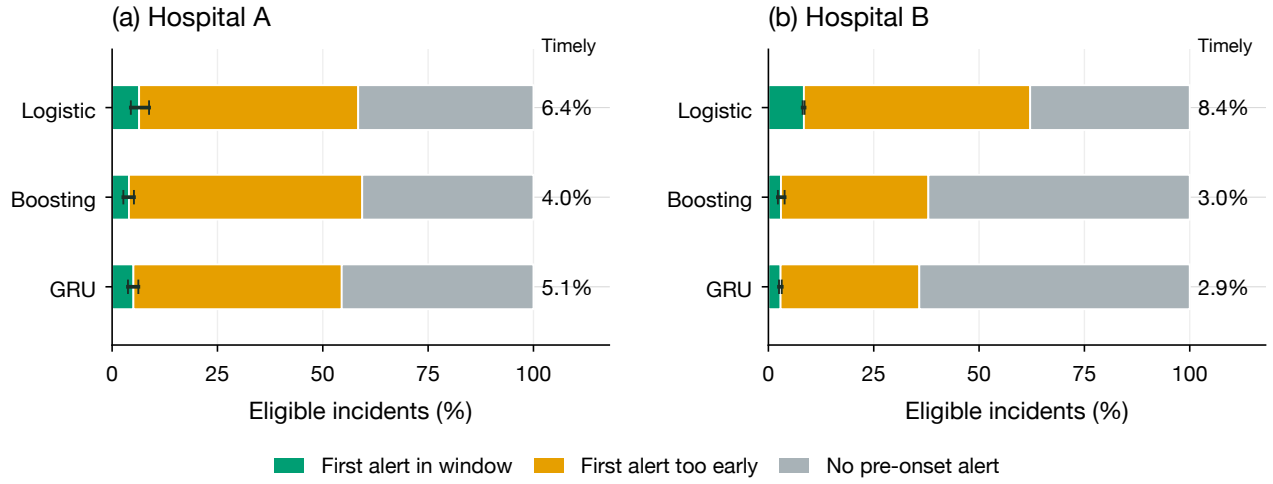
change does not establish preserved incident detection. The hospital-B GRU laboratory restriction is the clearest example: nearly unchanged AUROC coexists with a substantial increase in missed incidents at the original operating threshold.

This motivates a benchmark contribution centered on the evaluation design. Fixed patients, predictions generated by the same fitted model, explicit permission interventions and paired uncertainty allow another method to be compared on the same question. Reporting source availability, ranking, missed incidents and review burden exposes tradeoffs that an AUROC-only benchmark would conceal. The present release establishes reference baselines and a reproducible stress test; it does not establish that these baselines represent the best achievable performance under restricted data. Models trained with permission variation, source-specific architectures and validation-only adaptation are natural additional benchmark entries.

The broader research question also asks whether sensitive information is used with permission and whether conse-



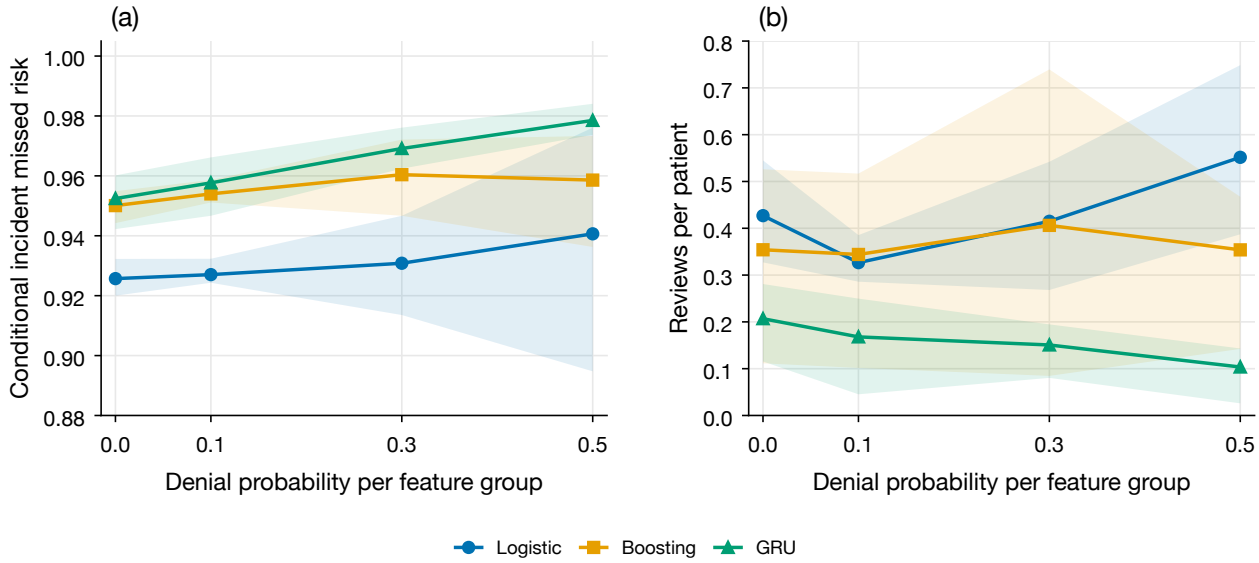
The tested first-alert grids remain far from the target region. All three seeds are shown (solid, dashed and dotted), with faint markers for dominated thresholds. Each model has 51 validation-quantile candidates plus endpoints before deduplication. Lines are frontier guides, not attainable interpolations. Shading marks risk ≤ 0.05 and workload ≤ 0.5. Panels show hospitals A and B; each inset enlarges risk 0.89–1.0 while the outer axes retain the complete risk range.



First-alert failures come from both early reviews and absent reviews. Bars show mean fractions across three seed-specific test sets at validation-utility-selected thresholds. Dark whiskers span the minimum and maximum timely fraction across seeds; they are descriptive ranges, not confidence intervals. Each label at the right gives the mean timely fraction.

quential actions receive effective oversight. The graph representation makes specified permissions and review paths testable. The statistical construction evaluates joint risk and workload on a frozen grid. The policy evaluation produces a negative answer for the tested first-alert family: the declared incident-risk target is not achieved, despite useful discrimination and positive challenge utility. Actual clinician effectiveness and patient benefit were not measured. Risk tolerances and review budgets are technical targets, not ethical thresholds derived from patients or clinicians. Changes in simulated consent affect benchmark alert performance and modeled workload; they are not measured causal effects on mortality, treatment or wellbeing.

This is more informative than observing that a bound failed to certify a weak model. The local confidence calculations exclude every tested candidate, and the test fron-



Admission opt-out changes incident risk and review demand together. Hospital A results use independently denied lab and vital groups and scenario-specific calibration-selected empirical fallback thresholds. Lines are three-seed means; shading spans the observed seed range. Panel (a) uses an expanded high-risk scale, not a zero baseline. No point carries a certified consent-scenario risk guarantee.

Mean pre-onset AUROC across three matched patient splits. Denial probabilities apply independently to the lab and vital groups; revocation uses probability 0.3 beginning at hour 12. Model parameters and test populations are unchanged. Across-seed means are descriptive; pointwise paired intervals for every split are released in the comparison CSV and illustrated for source-group restrictions.

Model	Site	Full	10%	30%	50%	No labs	No vitals	Metadata	Revoke
Logistic	A	0.745	0.736	0.726	0.719	0.729	0.715	0.689	0.731
Logistic	B	0.666	0.662	0.647	0.634	0.639	0.640	0.604	0.652
Boosting	A	0.794	0.780	0.763	0.747	0.765	0.741	0.710	0.766
Boosting	B	0.731	0.716	0.688	0.662	0.689	0.636	0.602	0.693
GRU	A	0.778	0.758	0.726	0.687	0.734	0.697	0.549	0.717
GRU	B	0.752	0.741	0.712	0.683	0.745	0.657	0.526	0.698

tiers remain far above the 5% risk target. The calibration pools are large enough that a zero-miss candidate could pass the finite-sample correction. The principal obstacle in this experiment is the combination of score trajectories and the definition of timely first review. More calibration data alone cannot turn a high population-risk candidate into a low-risk one.

The distinction also explains why the stronger predictors do not dominate the linear model on every endpoint. AUROC rewards correct ordering of individual hourly scores. Timely first review depends on the joint trajectory: scores before the window must remain below threshold, while at least one in-window score must cross it. The joint law of the pre-window and in-window maxima is therefore central. An improvement in hourly ranking need not improve that joint event, and a model that sends an early alert can receive utility credit while failing the strict first-review criterion.

Threshold nesting is often invoked to justify risk calibration, but the decision procedure sits between the score and the loss. Alert suppression, cooldown, a one-review limit or a persistent clinical assessment can change the loss family even when the score thresholds are unchanged. The counterexample in the first-alert non-monotonicity proposition isolates this issue without relying on a particular dataset or model architecture.

A useful operational policy may allow an early assessment to remain active, request a second review after deterioration, or suppress repeated notifications for a fixed period. None of these policies is ruled out by our results. Each would need a precise definition of review cost, the duration of an assessment and the event that counts as a successful action. The finite-grid method can evaluate non-monotone policies once their patient-level losses and workloads are specified, although a workload variable above one would require a bounded-mean or count-aware interval rather than the Bernoulli interval used here.

The distinction between infeasible and undetermined should remain visible in such evaluations. A confidence method that only returns accepted candidates cannot, by itself, distinguish an inadequate predictor from an underpowered calibration set. Our lower-bound exclusion criterion supplies that additional information, while limiting the conclusion to the actual finite grid. Extending an exclusion claim to all real-valued thresholds would require additional

structure or a uniform result beyond the present analysis.

The graph analysis and queue simulation answer complementary questions. The former asks whether a consequential action can be reached without traversing a review node. The latter asks whether a request receives a timely correct review under explicit assumptions. A path constraint can be satisfied even when every reviewer is overloaded or mistaken; a favorable queue result can coexist with an unmodeled bypass in a real software system.

The queue results are dominated by the upstream incident policy because a reviewer cannot repair a missing or mistimed request within the specified endpoint. This does not imply that reviewer quality is generally unimportant. It shows that improving one component cannot compensate for an event that the system never routes to that component in time. Clinical implementation evidence such as TREWS adoption (Henry et al., 2022) is needed to determine how review actually occurs, and externally validated alert-burden analyses such as Wong et al. (2021) are needed to assess the workload that a hospital would experience.

Permission changes alter more than the number of visible columns. A causal summary can retain earlier measurements; observation counts can reveal the measurement process; a recurrent state can encode information from a source that is no longer permitted. Clearing these paths is part of the implemented consent intervention. The conservative recurrent reset illustrates a real design tension: strict removal of mixed historical state can also remove useful permitted context.

The population consequences depend on who opts out, which sources are restricted and when permissions change. Our independent per-group denial process is a controlled stress test, not a model of real consent behavior. It is particularly important not to read the resulting cost curves as evidence that patients should disclose a particular type of information. Dynamic consent includes communication and participant agency (Kaye et al., 2014); predictive cost is only one input to that broader process.

The exchangeability statement clarifies the statistical boundary. A stable joint distribution of trajectories and permission states supports the certificate after the same frozen transformation is applied. Conditional outcome-ignorability of consent does not, by itself, make a changed deployment mechanism equivalent to the calibration mechanism. Hospital shift and consent shift can also occur together. Estimating a domain ratio from an initial feature vector cannot establish that the complete longitudinal patient, outcome and consent distribution has been correctly transported.

Limitations and next evaluation

The warning window is deliberately strict and retrospective. An alert just before its start counts as a failure even if an earlier assessment would benefit a patient. Onset is reconstructed from the supplied label rather than independently adjudicated, and workload stops at a retrospectively defined time. The study consequently evaluates a well-defined benchmark requirement; it does not establish an optimal clinical timing rule. A prospective design should specify an observable review episode and account for the persistence of an assessment.

The model comparison is limited to three implementations and modest tuning budgets. It excludes dedicated missingness-decay models, ensembles, richer temporal features and many possible training objectives. Class weighting affects probability calibration, and no post-hoc probability calibrator is fitted. Models adapted during training to restricted inputs have not been evaluated, so the results are not a lower bound on the achievable performance of consent-constrained AI.

The sampling assumptions are also stronger than the available clinical record can verify. We treat benchmark patient files as independent units. Repeated patients across hospitals, temporal changes in care, correlated ward practices and label error could violate a deployment interpretation. Bootstrap intervals condition on fitted models and fixed splits; the three seeds reuse many patients. Group checks use a small set of recorded variables and do not establish broader fairness. The calibration-subset draws overlap, and no certified rule exists for a conditional test-risk exceedance distribution.

Consent and reviewers are simulated. There are no observed preference trajectories, clinician assessments, service-time measurements or prospective patient outcomes. Arrival and fatigue models are chosen sensitivity functions, not fitted behavioral laws. The graph is a model of information flow rather than an audit of a deployed EHR integration. Consent masks do not provide differential privacy, cybersecurity protection, deletion from previously trained model parameters or a complete account of institutional governance.

The most direct next benchmark extension is a comparison against models trained on permitted inputs, including missingness-aware temporal models and training with heterogeneous source availability. Training permission and inference permission should be varied separately. A second clinical task would test whether the observed ranking-versus-operating-point discrepancy is specific to sepsis. Real preference data would be needed to replace the independent denial mechanism with a defensible model of participant choices.

Continuing-review and cooldown policies should also be evaluated using workload units that reflect actual assessment episodes. Prospective human evaluation would then need to measure alert interpretation, correct action, response time, overrides and patient outcomes. These are required measurements, not outcomes supplied by the present simulation.

Conclusion

Consent-related restrictions can be implemented and their predictive costs measured, but the tested sepsis systems do not preserve all aspects of performance when inputs change. The cost is source-, model- and site-dependent, and similar AUROC can coexist with substantially more missed incidents at a fixed threshold. Following hourly trajectories and clearing state at withdrawal connect the information constraint to its timing consequences. The benchmark therefore evaluates useful prediction and operating behavior together, while keeping permission enforcement, statistical guarantees and actual clinical oversight distinct. It provides evidence for comparing designs under explicit constraints; it does not decide which constraints patients should accept or how much clinical authority an AI system should receive.

Reproducibility and implementation checks

The public repository contains locked Python dependencies, exact data manifests, patient-level split assignments, fitted model outputs, frozen prediction arrays, tuning records and all result tables. The main full-release run used commit `fbfaff0d`; the frozen-policy audit used `652de9e8`. Exact source archives preserve both computations. The paired consent comparison uses commit `b270a1cb`, reuses the nine saved predictors, checks their full-input scores against the parent prediction arrays, and records hashes for the models and prediction inputs. Its configuration is frozen before evaluation. The publication instructions distinguish the full-model stage, the frozen-policy audit and this paired inference-only comparison; none requires the original partial-run resume.

The main evaluation ran on an Apple M4 CPU with 16 GiB memory and two numerical-library/torch threads. The completed full-data run took 9 minutes 3 seconds. It reused a completed logistic seed-11 fit from a 2 minute 40 second partial run that failed during constant-feature binning for boosting. Hashes verify the reused outputs. The later policy audit took 1 minute 20 seconds and did not retrain predictors. The paired consent run took 9 minutes 10 seconds end to end (508 seconds in the comparison stage) and did not retrain predictors. Peak memory was not instrumented. Earlier exploratory studies are archived separately and are not included in these run times.

The corrected reviewer audit changes the false-action eligibility rule: a patient with any positive label is not treated as a true negative merely because the patient is ineligible for the strict incident endpoint. The original queue output is retained for provenance and labeled superseded. Predictor scores, incident-loss calculations and calibration-certificate outcomes are unchanged by that correction.

Twenty-six implementation checks pass. These include causal feature invariance, consent/purpose masks, cache revocation, review-path equivalence, the first-alert non-monotonicity counterexample, exact binomial boundary cases, the distinction between infeasibility and insufficient evidence, agreement with official utility, handling of constant/all-missing features and false-action eligibility, paired zero-difference intervals and preservation of permitted values when denied histories are cleared. These checks are mechanised evidence about the implementation. The proofs in the mathematical analysis establish the mathematical results independently of those tests.

The primary public data retain their CC BY 4.0 license and the official evaluator retains its BSD license. No access-controlled dataset agreement is accepted by the code. No new patient or clinician recruitment occurred. This draft does not assert a project-specific institutional approval or exemption; applicable institutional requirements should be resolved before submission or any prospective study.

AI assistance was used in implementation, mathematical drafting, analysis and manuscript preparation. No LLM is used as a predictor or reviewer in the evaluated pipeline. Author responsibility includes checking the mathematical arguments, source attributions and interpretation before submission. The code, data, paper source and interactive results are available at `https://github.com/aravinds-kannappan/Clinical-AI` and `https://consent-clinical-ai.vercel.app`.

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